No. of pages: 03

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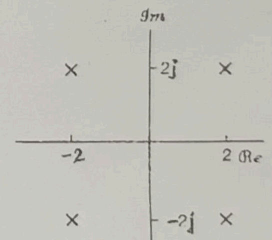
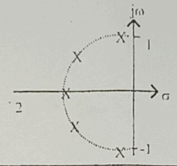
Roll No. 2024UEC25[redacted]

III SEMESTER B.Tech (ECE, EVDT, EIOT)
END-SEMESTER EXAMINATION NOVEMBER 2025

Course Code- ECECC302 / VTECC302/EIECC302/ ECECC05 / EIECC05
Course Title- Signals & Systems
Time: 3:00 Hrs Max Marks: 50

Note: - *Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.*

Q.N.		Questions	Marks	CO
Q1		**Attempt any 2 out of 3 parts**		
	a	Let two functions be defined by i $x(t) = \begin{cases} 1, & \sin(20\pi t) \geq 0 \\ -1, & \sin(20\pi t) < 0 \end{cases}$ and ii $x_2(t) = \begin{cases} t, & \sin(20\pi t) \geq 0 \\ -t, & \sin(20\pi t) < 0 \end{cases}$ Plot these two functions. Are they periodic? If yes, what are the time periods?	05	CO1
	b	Evaluate the following expressions: (i) $\int_{-\infty}^{\infty} \left[\cos\frac{\pi}{2}(t-5)\right].\delta(2t-3)dt$ (ii) $e^{2t}\cos\left(\frac{50}{\pi}t\right)\delta(t+1)$ (iii) Compute the Energy/Power of $x[n] = \cos\left[\frac{\pi n}{4}\right] + \sin\left[\frac{\pi n}{8}\right] - 2\cos\left[\frac{\pi n}{2}\right]$	05	
	c	Consider a discrete-time system with input $x[n]$ and $y[n]$ related by $y[n] = \sum_{k=n-n_0}^{n+n_0} x[k]$, Where n_0 is a finite positive integer. i Is this system linear? ii Is this system time-invariant? iii Is this system causal? iv Is this system stable?	05	
Q2		**Attempt any 2 out of 3 parts**		
	a	Which continuous time signal with fundamental time period 'T', will have the following Fourier series coefficients i $X_k = \frac{\sin^2(k\pi/2)}{4(k\pi/2)}$. ii $X_k = 2$ Plot the signal.	05	CO2
	b	If $f(t) * g(t) = c(t)$, then express i $f(at) * g(at)$ ii $f(-t) * g(-t)$ iii $f(t-t_0) * g(t+t_1)$ In terms of c(t).	05	
	c	Following information are given about a periodic signal $x[n]$ with fundamental period N = 8 and Fourier series coefficient X_k: i $X_k = -X_{k-4}$ ii $y[n] = \left(\frac{(1)^n + (-1)^n}{2}\right) x[n-1]$ Find the Fourier series Coefficients of $y(n)$.	05	
Q3		**Attempt any 2 out of 3 parts**		
	a	Find the following numerical values. i $X(f)\vert_{f=1/2}$, *when* $x(t) = rect(t) * rect(2t)$ ii $x(0)$, *when* $X(j\omega) = 2sinc(\omega/2\pi) \star 3sinc(\omega/\pi)$	05	CO3
	b	Find an expression for $y(t)$ using Fourier Transform. Do not use convolution. i $y(t) = e^{-t}(t)u(t) * \cos(\pi t)$ ii $y(t) = sinc(t) * sinc^2(t/2)$	05	
	c	Suppose we want to design a discrete-time LTI system which has the property that if the input is $x[n] = \left(\frac{1}{2}\right)^n u[n] - \frac{1}{4}\left(\frac{1}{2}\right)^{n-1} u[n-1]$ then the output is $y[n] = \left(\frac{1}{3}\right)^n u[n]$ (i) Find the impulse response and frequency response of a discrete time LTI system that has the foregoing property (ii) Is the system stable, causal.	05	
Q4		**Attempt any 4 out of 6 parts**		
	a	For the below given pole zero plotof X(s), i. Determine x(t) if $x(t)e^{-3t}$ is absolutely integrable. ii. Determine x(t) if $x(t) * (e^{-t}u(t))$ is absolutely integrable (Here * means convolution). 	2.5+ 2.5	CO4

	b	We are given the following five facts sbout a real signal $x(t)$ with Laplace transform $X(s)$: 1. $X(s)$ has exactly two poles. 2. $X(s)$ has no zeros in the finite s-plane. 3. $X(s)$ has a pole at $s = -1 + j$. 4. $e^{2t}x(t)$ is not absolutely integrable. 5. $X(0) = 8$ Determine $X(s)$ and Compue x(t).	3+2	
	c	X(s) has 5 poles at $s = -1,\ s = \sqrt{\frac{1}{2}} \pm j\sqrt{\frac{1}{2}},\ s = \pm j$ i Is the signal x(t) real, even, oscillatory? ii Find all the possible two sided functions x(t). 	02+ 03	
Q5		**Attempt any 2 out of 3 parts**		
	a	State and prove all the properties of ROC of Z Transform.	05	
	b	Find the inverse z-transform of this function in closed form using partial-fraction expansion, i $X(z) = \frac{z^2}{z^2 - z + 1/4}, \quad \lvert z\rvert < 0.5$ ii $X(z) = \frac{1}{z^2 + 0.8z + 0.16}, \quad \lvert z\rvert > 0.4$	05	CO5
	c	The transfer function of a system is given as $H(z) = \frac{z + 0.5}{(z + 0.4)(z - 2)}$ Specify the ROC of $H(z)$ and determine $h(n)$ for the following conditions: (i) The system is causal. (ii) The system is stable (iii) Can the given system be both causal and stable?	05	